GUANLIN DEREK HUANG

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West Lafayette, Indiana

Research interests: robot learning, visuotactile manipulation, world models, imitation learning

EDUCATION

Purdue University, West Lafayette, Indiana **Expected May 2027**

Bachelor of Science in Industrial Engineering

- Relevant coursework: optimization, stochastic models, control and networks, Python data analysis, linear algebra, differential equations, digital systems, and advanced C programming.

RESEARCH EXPERIENCE

Mechanisms and Robotic Systems Lab, Purdue University **Oct 2025 - Present**

Undergraduate Research Assistant | Advisor: Prof. Yu She

- Contribute to Atomic Policy, a research framework that decomposes long horizon manipulation into task planning, visual grounding, action-specific Diffusion Policy skills, and execution checking; supporting an ongoing conference manuscript.
- Developed SmartData, an open-source Python pipeline that converts iPhone mounted handheld gripper recordings (RGB, depth visualization, and 6 DoF pose) into synchronized, training ready robotic manipulation datasets.
- Integrated SAM 2 gripper segmentation, hindsight contact labeling, Moondream 2 and MolmoPoint object grounding, Rerun diagnostics, and LeRobot/Zarr export in a reproducible one command workflow.
- Designed a geometry aware extension for persistent 3D target tracking using calibrated camera projection, gated SAM 2 refinement, failure detection, and re-grounding for future wrist camera deployment.
- Developed and tested UR5 and Franka Panda control workflows in MuJoCo and on hardware; collaborated with PhD researchers on dexterous hand and VLA/VLM manipulation policy development.

Integrated Intelligent Electric Power Grid Lab, Purdue University **Apr 2025 - Oct 2025**

Research Assistant | Midwest Power Grid Capacity Expansion Study | Advisor: Prof. Andrew Liu

- Developed inputs for the SWITCH linear programming capacity expansion model to study MISO resource adequacy under decarbonization scenarios.
- Built stochastic renewable generation and load profiles across multiple weather years and modeled fossil retirements, renewable expansion, capacity adequacy, and firm capacity requirements.
- Performed scenario and time series analyses of system cost and reliability under renewable penetration, emerging loads, storage, and demand-side flexibility.

ADDITIONAL RESEARCH AND EXPERIENCE

Machine Learning in Medical Imaging and Computational Biology **Jun 2021 - Aug 2021**

Research Program | MIT | Advisor: Prof. Manolis Kellis

- Applied convolutional neural networks, semantic segmentation, and one shot learning to liver cancer histopathology classification and gene mutation prediction.
- Coauthored a team research paper and presented the findings to Prof. Kellis.

Innogrit **May 2024 - Aug 2024**

NAND Development and Validation Intern

- Executed chip-validation tests and analyzed NAND endurance, latency, and error behavior to identify functional and performance issues during early development.
- Debugged hardware communication and test workflows over I2C and SMBus; analyzed results and documented findings for design iteration.
- Used Python, C/C++, Verilog, Linux, and Excel for test automation, data analysis, and reporting.

SELECTED ROBOTICS PROJECT

General Electric Robotics and Autonomous Systems **Jan 2024 - May 2024**
Vertically Integrated Project
Purdue University

- Implemented forward and inverse kinematics, Jacobian-transpose force mapping, and an iterative pseudoinverse IK solver with convergence and singularity handling for a two-link robotic arm.
- Designed a Cartesian-space PD controller for nonlinear trajectory tracking and converted task-space trajectories into joint commands.
- Validated the control pipeline in MuJoCo using trajectory visualization and mean-squared error, extending the system toward pick-and-place and motion planning.

LEADERSHIP

Purdue Chinese Students and Scholars Association **Aug 2025 – Jun 2026**
Event Planning Secretary and Vice Chair, Career Development

- Coordinate event logistics, scheduling, permits, promotion, vendor communication, alumni outreach, and company information sessions for Purdue students.

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, Java, R, Verilog, LaTeX
Robotics and ML: MuJoCo, SAM 2, Rerun, LeRobot, Zarr, Diffusion Policy, VLA/VLM, kinematics, trajectory control
Systems and Data: Linux, Git, GitHub, OpenCV, Excel, multimodal data pipelines, time series and scenario analysis
Languages: English and Mandarin (native)